

MATH 116: Practice Set 3

This assignment is for practice only and is not to be submitted.

It will be helpful in preparing for Tutorial 3.

1. Let $h(x)$ be the Heaviside function, defined as:

$$h(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x > 0 \end{cases}.$$

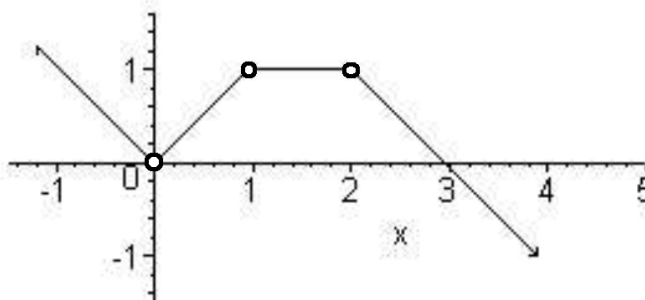
- (a) Write the function

$$f(x) = x + (1 - x)h(x - 1) + (4x - x^2 - 4)h(x - 2)$$

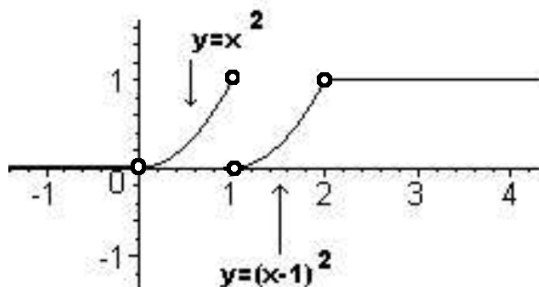
in piecewise-defined form and sketch it.

- (b) Consider the two graphs shown. In each case, express the function shown in terms of Heaviside functions (you may wish to write it in piecewise-defined form first, although this isn't necessary)

i.



ii.



2. Rewrite the following expressions in the form of a single *general sine function* (of the form $A \sin(\theta + \varphi)$):

(a) $4 \cos \theta - 4 \sin \theta$

(b) $\sin \theta + 8 \cos \theta$

3. Find all $\theta \in [0, 2\pi]$ such that $\sqrt{3} \sin \theta - \cos \theta = 2$.

Hint: Start by rewriting $\sqrt{3} \sin \theta - \cos \theta$ in the form $A \sin(\theta + \varphi)$.